Blog Notes 3

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June 5, 2020

Introduction

- Stochastic Differential Equation
- Stochastic Calculus V.S. Real Calculus
- Itô Formula
- Brownian Motion
- Geometric Brownian Motion
- Hull-White Interest Rate model

Notes:

In this note, I would write my understanding of Solving stochastic integral, using Itô Formula and how to apply it to solve three SDE examples(Brownian Motion, Geometric Brownian Motion, one Interest Rate model).

Stochastic Differential Equation

A generic form of a stochastic differential equation is given by

$$dX(t) = \beta(t, X(t))dt + \gamma(t, X(t))dW(t)$$

where,

- $\beta(t, X(t))$ is called a drift,
- $\gamma(t, X(t))$ is called a diffusion.
- $W(t)$ is a Brownian motion.

Stochastic Differential Equation

Examples

- Dynamics of a Brownian Motion is $dX(t) = \mu dt + \sigma dW(t)$, where $\beta(t, X(t)) = \mu, \gamma(t, X(t)) = \sigma$.

- Dynamics of a Geometric Brownian Motion(GBM) is $dX(t) = \mu X(t)dt + \sigma X(t)dW(t)$. where $\beta(t, X(t)) = \mu X(t), \gamma(t, X(t)) = \sigma X(t)$.
- The Dynamics of a GBM can be written as $\frac{dX(t)}{dt} = \mu dt + \sigma dW(t)$, and can be observed that it can model the Dynamics of Returns on an Asset Price $X(t)$.

- Dynamics of a Hull-White Interest model is $dR(t) = (a(t) - b(t) \cdot R(t))dt + \sigma(t) \cdot d\widetilde{W}(t)$ where $\beta(t, R(t)) = a(t) - b(t) \cdot R(t), \gamma(t, R(t)) = \sigma(t)$.

Stochastic Differential Equation

When given a stochastic differential equation which models dynamics of a stochastic process, in generic form as follows,

$$dX(t) = \beta(t, X(t))dt + \gamma(t, X(t))dW(t)$$

Together with an given initial condition as

$$X(t) = x, \quad t \geq 0 \quad \text{and} \quad x \in \mathbb{R}$$

Question: What is solving the stochastic differential equation?

Answer: Solving the SDE is to solve for a function $X(T)$, which is a stochastic process that satisfies the SDE and the initial condition.

Idea: Integrate both sides, from time t to time T;

$$\int_t^T dX(s) = \int_t^T \beta(s, X(s))ds + \int_t^T \gamma(s, X(s))dW(s)$$

Stochastic Calculus V.S. Real Calculus

Solving stochastic differential equation involves integrals with respect to stochastic process and integrals with respect to time, so it involves stochastic calculus and real calculus.

Example:

take the Itô integral

$$\int_t^T dW(s), \text{ where, } W(t) \text{ is a standard Brownian motion}$$

as an example to see the difference between stochastic calculus and real calculus.

Stochastic Calculus V.S. Real Calculus

Two ways to explain what the stochastic integral $\int_t^T dW(s)$.

1. $\int_t^T dW(s)$ means adding up all the little changes $dW(s)$ from time t to time T.

   Assume a small time interval Δ, sum up all the changes of $W(s)$ from t to T in the small consecutive intervals and then make the time interval smaller and smaller. That is,

$$
\begin{aligned}
& \int_t^T dW(s) \\
= \quad & \lim_{\Delta \to 0} [(W_{t+\Delta} - W_t) + (W_{t+2\Delta} - W_{t+\Delta}) + \ldots + (W_T - W_{T-\Delta})] \\
= \quad & \lim_{\Delta \to 0} (W_T - W_t) = W(T) - W(t)
\end{aligned}
$$

Stochastic Calculus V.S. Real Calculus Continue

Corresponding to the stochastic integral $\int_t^T dW(s)$, in real calculus, an integral interested is $\int_t^T \frac{dW(s)}{ds} ds$.

In real calculus, the integral means calculating the area under the curve chopped in the time interval dt.

Because Brownian motion $W(t), t \geq 0$ has rough(non-smooth) trajectories with respect to time and it fluctuates a lot in short period, there is no smooth function under which the area can be taken.

Stochastic Calculus V.S. Real Calculus Continue

To calculate the integral of the non-smooth (non-differential) curve $W(t)$, use the notation $dW(t)$.

In real calculus, the conventional notation for the integral with respect to time is typically for smooth function.

However, **in stochastic calculus**, the idea is adding up all the little changes of the process, $dW(t)$, and get the level of the change of the process $W(t)$ from time t to time T, that is,

$$\int_t^T dW(s) = W(T) - W(t)$$

Stochastic Calculus V.S. Real Calculus Continue

There is another way to explain what the stochastic integral $\int_t^T dW(s)$.

2. $\int_t^T dW(s) = W(T) - W(t)$ is a random variable.

   By the definition of Brownian motion, its increment $W(T) - W(t)$ is a normally distributed random variable with

$$\begin{cases} \text{Mean} & = & E(W_T - W_t) = 0 \\ \text{Variance} & = & Var(W_T - W_t) = T - t \end{cases}$$

   That is,

$$W_T - W_t = N(0, T - t) = \sqrt{(T - t)}N(0, 1)$$

   Actually,

$$dW(t) = \sqrt{dt}, \quad (dW(t))^2 = dt$$

Stochastic Calculus

Principal to Keep in Mind:

The basic principal to keep in mind is that **stochastic calculus** is like a regular calculus, but it do second order Taylor expansions and keep the term dt, $dW(t)$ and $(dW(t))^2$ ($(dW(t))^2$ is the same as dt term, because $(dW(t))^2 = dt$), omit the higher order terms.

In contrast, **real calculus** do the first order Taylor expansions, only keep dt term and don't keep higher order terms.

Basic Lemma to use in Itô formula:

In the Itô formula, take the differential to second order expansions and omit the terms $dtdW(t)$, $dtdt$, simplify the term $dW(t)dW(t)$, by letting

$$\begin{cases} dtdW(t) &= 0, \quad dtdt = (dt)^2 = 0 \\ dW(t)dW(t) &= (dW(t))^2 = dt \end{cases}$$

Itô Formula

An important multivariate version Itô Formula

In stochastic calculus, suppose $\{X_t, t \geq 0\}$, $\{Y_t, t \geq 0\}$ are two Itô process, for a C^2-function $f(x, y)$ which is twice continuously differentiable function, then, the composition of $f(x, y)$ applied to $X(t)$ and $Y(t)$ is again an Itô process, with the differential given by

$$\begin{aligned} & df(X_t, Y_t) \\ = & \frac{\partial f}{\partial X} dX + \frac{\partial f}{\partial Y} dY + \frac{\partial^2 f}{\partial X^2} dXdX + \frac{\partial^2 f}{\partial Y^2} dYdY + \frac{\partial^2 f}{\partial X \partial Y} dXdY \\ = & f_X dX + f_X dX + \frac{1}{2} f_{X,X} (dX)^2 + \frac{1}{2} f_{Y,Y} (dY)^2 + f_{X,Y} dXdY \end{aligned}$$

In the above formula, use simplified notation dX, dY representing $dX(t)$, $dY(t)$. It is noted that the term $\frac{\partial^2 f}{\partial X \partial Y} dXdY$ is from $\frac{1}{2} \frac{\partial^2 f}{\partial X \partial Y} dXdY + \frac{1}{2} \frac{\partial^2 f}{\partial Y \partial X} dYdX$, where $\frac{\partial^2 f}{\partial X \partial Y} dXdY = \frac{\partial^2 f}{\partial Y \partial X} dYdX$.

Itô Formula Continue

In the two variable version of Itô Formula, $\{X_t, t \geq 0\}$, $\{Y_t, t \geq 0\}$ are two Itô processes, in the differential of $df(X(t), Y(t))$, there are terms $dX(t)$, $dY(t)$, $dX(t)dX(t)$, $dY(t)dY(t)$, $dX(t)dY(t)$.

Following the basic lemma to use in the Itô formula, that is,

$$\begin{cases} dtdW(t) &= 0, \quad dtdt = (dt)^2 = 0 \\ dW(t)dW(t) &= (dW(t))^2 = dt \end{cases}$$

In the multivariate version Itô Formula, if
$dX(t) = \mu_1(t)dt + \sigma_1(t)dW(t)$ and
$dY(t) = \mu_2(t)dt + \sigma_2(t)dW(t)$, where both are decomposed to dt term and $dW(t)$ term,
then, **another lemma to simplify the formula** is

$$\begin{cases} dX(t)dX(t) &= (\sigma_1(t)dW(t))^2 = \sigma_1^2(t)dt, \\ dY(t)dY(t) &= (\sigma_2(t)dW(t))^2 = \sigma_2^2(t)dt, \\ dX(t)dY(t) &= \sigma_1(t)\sigma_2(t)(dW(t))^2 = \sigma_1(t)\sigma_2(t)dt \end{cases}$$

Common Formula Example in Stochastic Calculus:

In stochastic calculus, we have

$$df(X,Y) = f_X dX + f_Y dY + \frac{1}{2} f_{X,X}(dX)^2 + \frac{1}{2} f_{Y,Y}(dY)^2 + f_{X,Y} dX dY$$

Example1:

Suppose $f(X,Y) = X \cdot Y$, we have

$$d(XY) = YdX + XdY + dXdY$$

which follows the stochastic calculus formula $df(X,Y)$ and from

$$\begin{cases} f(X,Y) = XY, \\ f_X = Y, \quad f_Y = X, \\ f_{X,X} = 0, \quad f_{Y,Y} = 0, \quad f_{X,Y} = 1, \quad f_{Y,X} = 1 \end{cases}$$

Common Multivariate Itô Formula Examples:

In the Itô formula, the aim is to find a stochastic differential of a composition of C^2-function applied to Itô process,

Example2: two variable version Itô formula

Suppose the composition of a twice continuously differentiable function $f(x, y) = xy$ is applied to two Itô processes

$$dX_i(t) = \mu_i(t)dt + \sigma_i(t)dW(t), \quad i = 1, 2$$

The product $X_1(t)X_2(t)$ is again an Itô process.
Find: $d(X_1(t)X_2(t))$.
Answer:

$$d(X_1(t)X_2(t)) = X_1(t)dX_2(t) + X_2(t)dX_1(t) + \sigma_1(t)\sigma_2(t)dt$$

which follows the stochastic calculus formula $d(XY)$ and from

$$\begin{cases} d(XY) &= YdX + XdY + dXdY, \\ dX_1(t)dX_2(t) &= \sigma_1(t)\sigma_2(t)(dW(t))^2 = \sigma_1(t)\sigma_2(t)dt \end{cases}$$

Common Multivariate Itô Formula Examples:

Example 3: two variable version Itô formula, one variable is time t

Suppose $\{X(t), t \geq 0\}$ is an Itô process, its dynamic is modelled by $dX(t) = \mu(t)dt + \sigma(t)dW(t)$ and suppose an C^2-function $f(x, y)$.

Find: $d(f(t, X(t)))$.

Answer:

$$\begin{aligned} d(f(t, X(t))) &= f_t dt + f_X dX(t) + \frac{1}{2} f_{X,X} (dX(t))^2 \\ &= f_t dt + f_X dX(t) + \frac{1}{2} f_{X,X} \sigma^2(t) dt \end{aligned}$$

which follows the stochastic calculus formula $df(X, Y)$ and from

$$\begin{cases} (dt)^2 = 0, \quad dtdX(t) = 0 \\ \text{[The term } dtdX(t) \text{ contains } (dt)^2 = 0, dtdW(t) = 0] \\ dX(t)dX(t) = (\sigma(t)dW(t))^2 = \sigma^2(t)dt \end{cases}$$

Single variable version Itô formula

An important one variable version Itô formula

Suppose $\{X(t), t \geq 0\}$ is an Itô process, its dynamic is modelled by $dX(t) = \mu(t)dt + \sigma(t)dW(t)$ and suppose an C^2-function $f(x)$. Then, the composition of $f(x)$ applied to the Itô process $\{X(t), t \geq 0\}$ is again an Itô process with the differential given by

$$\begin{aligned} d(f(X(t)) &= f_X dX(t) + \frac{1}{2} f_{X,X} (dX(t))^2 \\ &= f_X dX(t) + \frac{1}{2} f_{X,X} \sigma^2(t) dt \end{aligned}$$

which follows the stochastic calculus formula $df(X, Y)$ and from

$$\begin{cases} f_t = 0, \quad f_{t,t} = 0 \quad f_{t,x} = f_{x,t} = 0 \\ \text{[No need to consider terms } dtdt, \ \ dtdX(t) or dX(t)dt] \\ dX(t)dX(t) = (\sigma(t)dW(t))^2 = \sigma^2(t)dt \end{cases}$$

Compared to Real Calculus

Note1: In real calculus, suppose a function $f(x)$ and a function $X(t)$, the composition of $f(x)$ applied to the function $X(t)$, the differential of $f(X(t))$ is $df(X(t)) = f_X dX(t)$ and this would only hold if $X(t)$ has differentiable trajectories.

2: In the one variable version Itô formula, recall that the term $\sigma(t)dW(t)$ relates to the stochastic integral $\int_t^T \sigma(t)dW(s)$, which has rough trajectories (not differentiable trajectories) and inherited by $X(t)$, and it causes the second order correction term. It also explains why this one variable version Itô formula only works for C^2-functions $f(x)$ (twice continuously differentialble function).

Compared to Real Calculus

Note3: Similarly, in real calculus, for multivariate function $f(x, y) = xy$ applied to two deterministic functions $X_1(t)$, $X_2(t)$. The differential is the product rule $dX_1(t)X_2(t) = X_1(t)dX_2(t) + X_2(t)dX_1(t)$, which is from the diffential of $f(x, y)$ in real calculus; $df(x, y) = f_y dy + f_x dx$.

It only holds for $X_1(t)$, $X_2(t)$ had no rough trajectories (differentiable trajectories). But due to the stochastic integral $\int_t^T \sigma_i(t)dW(s)$ inherited by $X_i(t)$, then we have the second order correction term.

Common single variable Itô Formula Examples:

Example 1: Exponential $f(x) = e^x$ applied to an Itô process

Suppose $\{X(t), t \geq 0\}$ is an Itô process, its dynamic is modelled by $dX(t) = \mu(t)dt + \sigma(t)dW(t)$.
An exponential function $f(x) = e^x$ is applied to the Itô process $\{X(t), t \geq 0\}$, denoted by $Y(t) = e^{X(t)}$.

Find: $dY(t)$.

Answer: Because $Y(t)$ is a smooth function (differentiable to any higher order, including order 2), we can use one variable Itô formula, the decomposition of $dY(t)$ is given by

Common single variable Itô Formula Examples:

$$
\begin{aligned}
d(Y(t)) &= d(e^{X(t)}) = f_X dX(t) + \frac{1}{2} f_{X,X} \sigma^2(t) dt \\
&= e^{X(t)} dX(t) + \frac{1}{2} e^{X(t)} \sigma^2(t) dt \\
&= Y(t) dX(t) + \frac{1}{2} Y(t) \sigma^2(t) dt
\end{aligned}
$$

which follows the stochastic calculus formula $df(X(t))$ and from

$$
\begin{cases}
f(x) = e^x, \quad f_x = e^x \;\; f_{x,x} = e^x \\
\\
df(X(t)) = f_X dX(t) + \frac{1}{2} f_{X,X} (dX(t))^2 \\
\qquad\qquad = f_X dX(t) + \frac{1}{2} f_{X,X} \sigma^2(t) dt
\end{cases}
$$

Common single variable Itô Formula Examples:

Definition

Motivated from the example, Suppose $\{X(t), t \geq 0\}$ is an Itô process, its dynamic is modelled by $dX(t) = \mu(t)dt + \sigma(t)dW(t)$.

there is a counter-part to the classical exponential function in stochastic calculus, which would have a same formula as the differential of exponential in real calculus. It is called the stochastic exponential.

Denoted by $\mathcal{E}(X(t))$, **the stochastic exponential** is defined by

$$\mathcal{E}(X(t)) = e^{X(t) - \frac{1}{2}\int_0^t \sigma^2(s)ds}$$

Fact:

Suppose $\{X(t), t \geq 0\}$ is an Itô process, its dynamic is modelled by $dX(t) = \mu(t)dt + \sigma(t)dW(t)$. $\mathcal{E}(X(t))$ satisfy the stochastic differential equation(SDE): $d\mathcal{E}(X(t)) = \mathcal{E}(X(t))dX(t)$

Common single variable Itô Formula Examples:

Motivation idea

The way to define the stochastic exponential is that we would like have the form $d(Y(t)) = d(e^{X(t)}) = e^{X(t)}dX(t)$ same as $d(e^{x(t)}) = e^{x(t)}dx(t)$, not appear the second order correction term $\frac{1}{2}Y(t)\sigma^2(t)dt$.

The way is to assume an exponential $Y(t)$ where after applying Itô formula, it has the decomposition

$$dY(t) = -\frac{1}{2}Y(t)\sigma^2(t)dt + Y(t)dX(t) + \frac{1}{2}Y(t)\sigma^2(t)dt$$

Common single variable Itô Formula Examples:

Motivation idea Continue

We know that if the exponential is $Y(t) = e^{X(t)}$, its decomposition is $Y(t)dX(t) + \frac{1}{2}Y(t)\sigma^2(t)dt$, our aim is to construct another form of exponential $Y(t)$ such that it has one more term $-\frac{1}{2}Y(t)\sigma^2(t)dt$ after using Itô formula.

The decomposition of the exponential differential has three terms, consider the Itô formula
$d(f(t, X(t))) = f_t dt + f_X dX(t) + f_{X,X}\sigma^2(t)dt$

We would like $f(t, X(t))$ be an exponential, compared to the formula $d(f(X(t))) = d(e^{X(t)}) = e^{X(t)}dX(t) + \frac{1}{2}e^{X(t)}\sigma^2 dt$

Then, we would be motivated to have the exponential defined by $f(t, X(t)) = e^{X(t) - \frac{1}{2}\int_0^t \sigma^2(s)ds}$, denoted the exponential by $\mathcal{E}(X(t))$, and call it **the stochastic exponential**.

Common single variable Itô Formula Examples:

Example 2: Check the Itô formula of the Exponential $f(t,x) = e^{x-\frac{1}{2}\int_0^t \sigma^2(s)ds}$ applied to an Itô process

Suppose $\{X(t), t \geq 0\}$ is an Itô process, its dynamic is modelled by $dX(t) = \mu(t)dt + \sigma(t)dW(t)$.
An exponential function $f(t,x) = e^{x-\frac{1}{2}\int_0^t \sigma^2(s)ds}$ is applied to the Itô process $\{X(t), t \geq 0\}$, denoted by
$Y(t) = f(t, X(t)) = e^{X(t)-\frac{1}{2}\int_0^t \sigma^2(s)ds}$.

Find: $dY(t)$.

Answer:
$f_t(t,X) = -\frac{1}{2}\sigma^2(t)Y(t)$, $f_X(t,X) = Y(t)$, $f_{X,X}(t,X) = Y(t)$

Apply the Itô formula $df(t, X(t)) =$
$f_t dt + f_X dX(t) + \frac{1}{2} f_{X,X}(dX(t))^2 = f_t dt + f_X dX(t) + \frac{1}{2} f_{X,X}\sigma^2(t)dt$,

We get: $dY(t) = Y(t)dX(t)$

Common single variable Itô Formula Examples:

Example 2: Logarithmic function $f(x) = log(x)$ applied to an Itô process

Suppose $\{X(t), t \geq 0\}$ is an Itô process, its dynamic is modelled by $dX(t) = \mu(t)dt + \sigma(t)dW(t)$.
An logarithmic function $f(x) = log(x)$ is applied to the Itô process $\{X(t), t \geq 0\}$, denoted by $f(X(t)) = logX(t)$.

Find: $d\ logX(t)$.

Answer:
$f_X(X) = \frac{1}{X(t)}$, $f_{X,X}(X) = \frac{-1}{(X(t))^2}$

Apply the Itô formula

$$df(t, X(t)) = f_X dX(t) + \frac{1}{2} f_{X,X}(dX(t))^2 = \frac{dX(t)}{X(t)} - \frac{1}{2}\left(\frac{dX(t)}{X(t)}\right)^2,$$

We get: $d\ logX(t) = \frac{dX_t}{X(t)} - \frac{1}{2}\left(\frac{dX(t)}{X(t)}\right)^2 = \frac{dX_t}{X_t} - \frac{1}{2}\left(\frac{dX_t}{X_t}\right)^2$